

Parallel Attention || MoE with Periodic FusionMoE

A controlled, pre-registered A / B / C experiment

~160M-active-parameter MoE language models · FineWeb-Edu · single A100 40GB

The question

Modern MoE LLMs (DeepSeek-V3, Kimi K2, GLM-4.5) are **serial**: the MoE reads the attention output of the same layer.

GPT-J and PaLM run the two sublayers **in parallel** from the same input.

1. Does a **parallel MoE block** lose quality?
2. Can a few **FusionMoE** layers (a token-wise MoE after every 4 parallel blocks) win it back?
3. Is the result a better **quality-per-compute** trade-off?

Three architectures, everything else identical

	block	MoE layers	expert width	FLOPs / token
A	$a = x + \text{Attn}(x) \rightarrow y = a + \text{MoE}(a)$	12	1920	1.068 G
B	$y = x + \text{Attn}(x) + \text{MoE}(x)$	12	1920	1.068 G
C_same	B + FusionMoE after blocks 4, 8, 12	15	1920	1.227 G (+15%)
C_matched	same, experts narrowed so $15 \times 1536 = 12 \times 1920$	15	1536	1.068 G

12 layers · d 768 · 8 SwiGLU experts, top-2 · same data, order, recipe, GPU · A and B start from **identical weights**

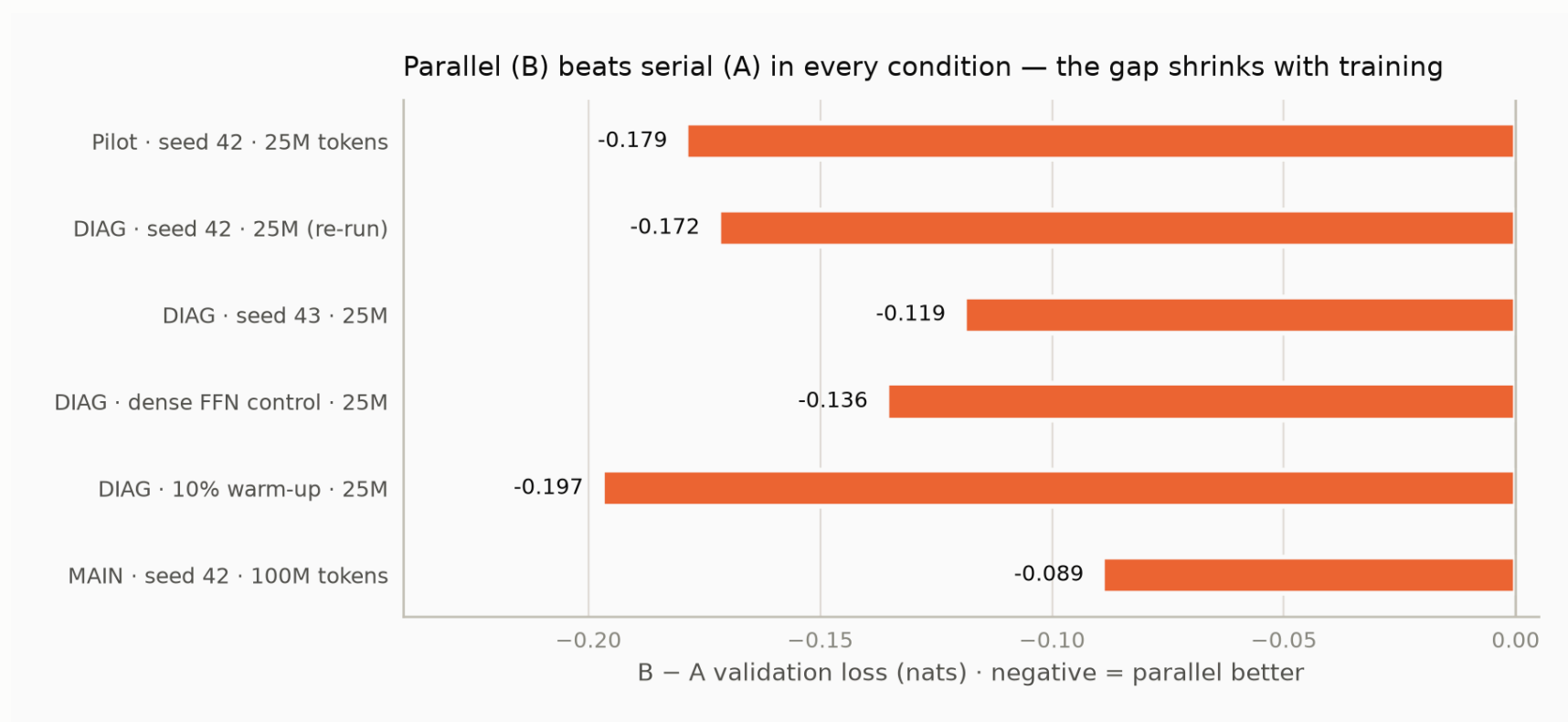
17 automated fairness checks after every run · decision rules written before each phase

Three phases

phase	tokens / model	what	outcome
Pilot	25M	A, B, C_same, C_matched	B beats A by 0.179 (unexpected)
Diagnostics	25M	A vs B × 4 conditions	the gap is real, not MoE-specific
MAIN	100M	all four	fusion verdict: inconclusive

Each phase's rules are a dated amendment in EXPERIMENT_SPEC.md (Amendments 1-5).

Diagnostics: is "parallel beats serial" real?

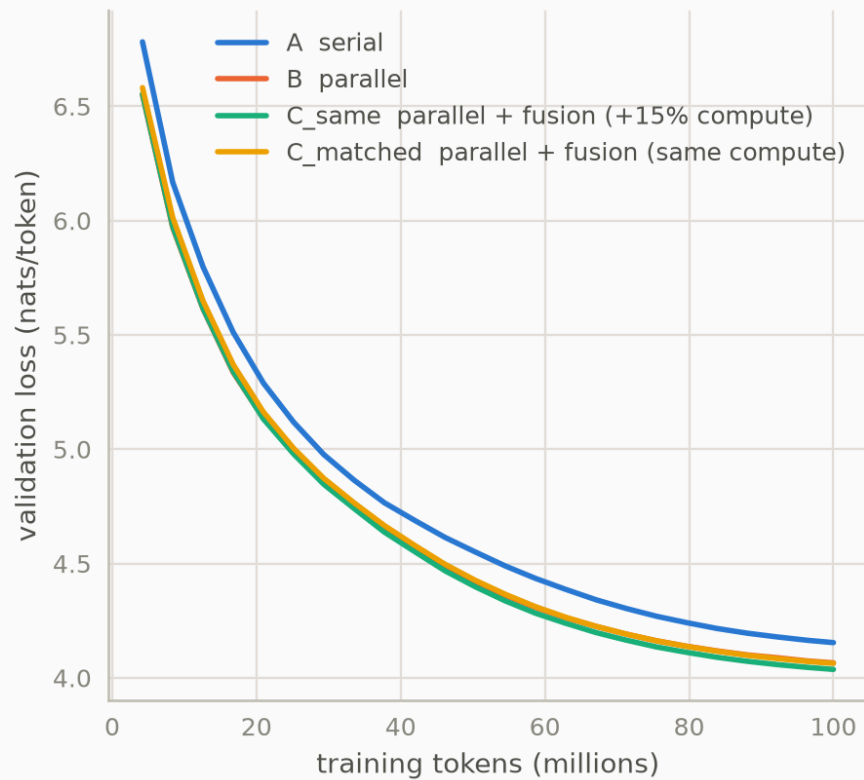


Robust across seeds · also with a **dense FFN** · not a warm-up artefact → a property of this **regime**, not of MoE routing.

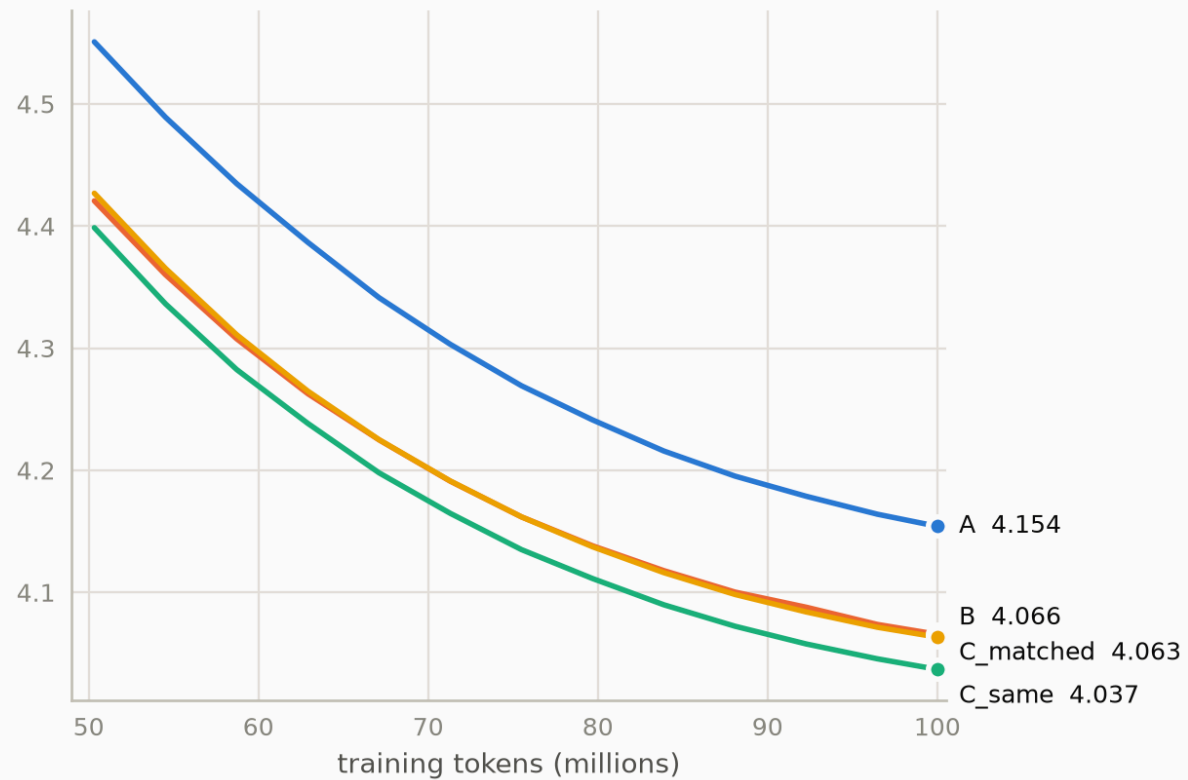
Seed-to-seed spread 0.053 nats → the single-seed decision threshold for MAIN.

MAIN: 100M tokens

Validation loss vs training tokens (MAIN, 100M tokens)



Zoom: second half of training

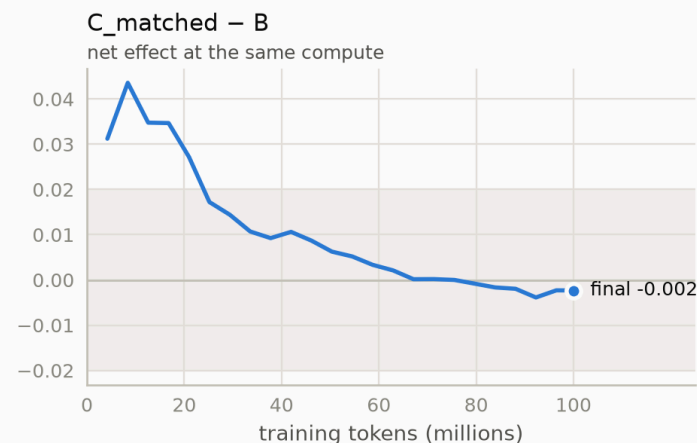
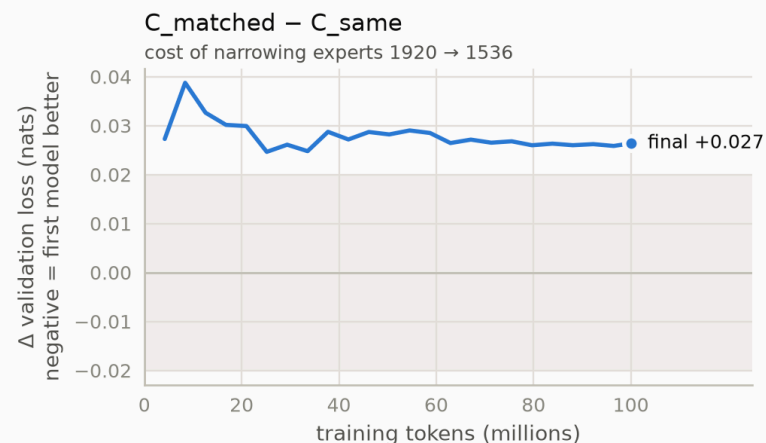
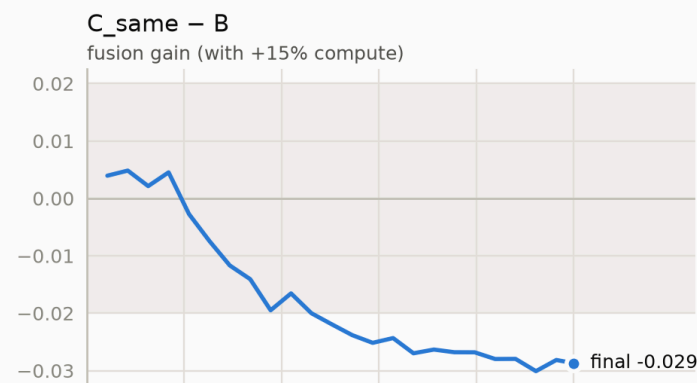
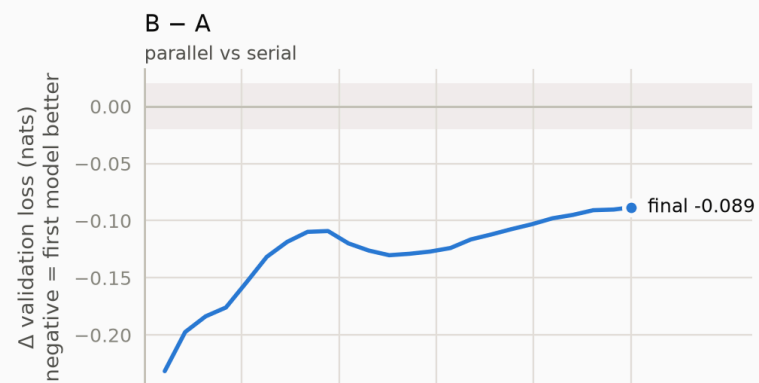


MAIN: final numbers (15,088 validation sequences)

model	val loss	ppl	time	tokens/s
A serial	4.1564	63.8	32.3 min	51.6k
B parallel	4.0672	58.4	32.3 min	51.6k
C_same	4.0379	56.7	37.7 min	44.2k
C_matched	4.0651	58.3	35.2 min	47.3k

paired ΔL	value [95% CI]	verdict
B – A	–0.089 [–0.090, –0.088]	better
C_same – B	–0.029 [–0.030, –0.029]	inconclusive (< 0.053)
C_matched – B	–0.002 [–0.003, –0.002]	equal

When does each effect appear?



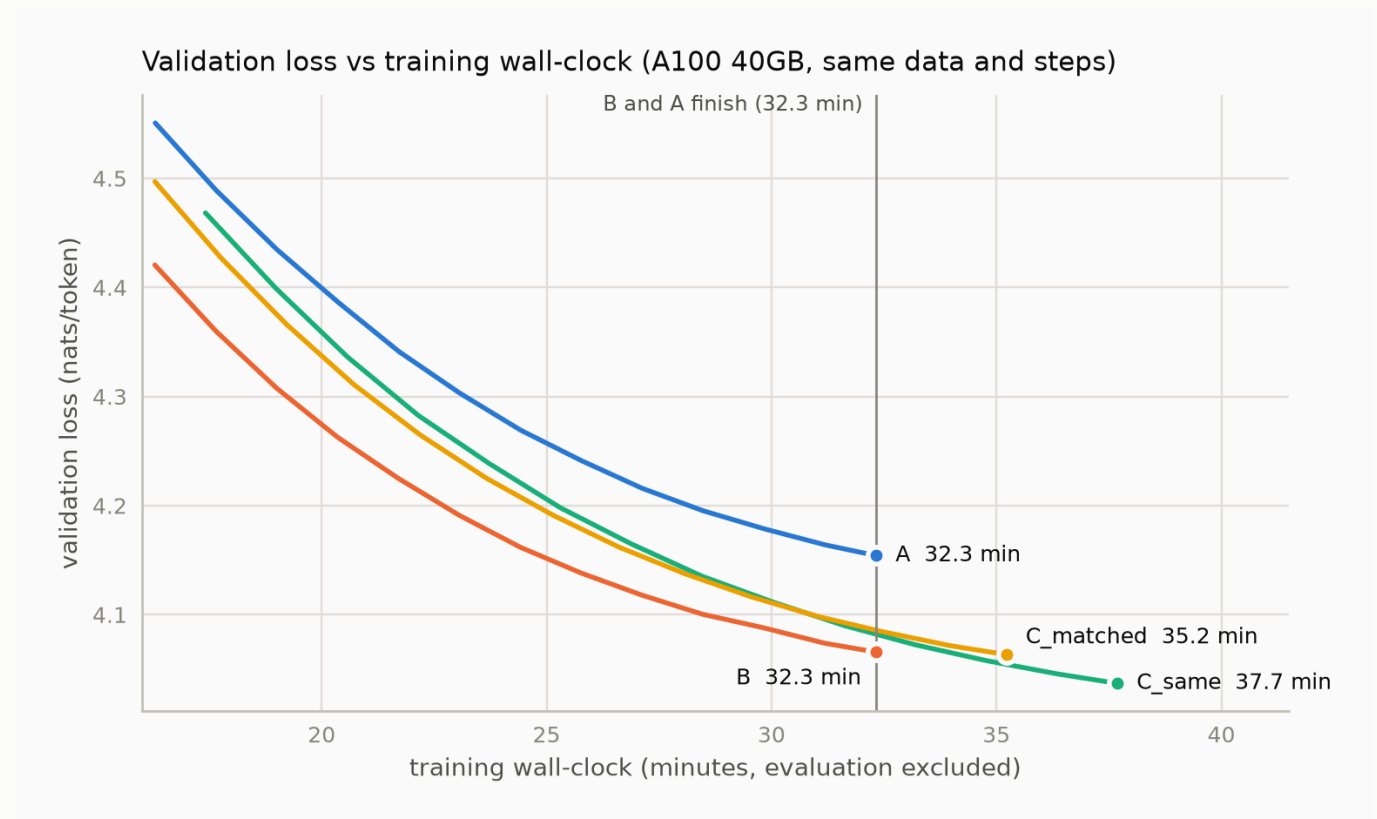
Paired differences on the same validation sequences. Shaded band: ± 0.02 -nat equivalence margin. Curves use the periodic 2,048-sequence subset; 'final' uses the full validation set (15,088 sequences).

Reading the gaps

- **B – A** shrinks $0.23 \rightarrow 0.089$: parallel learns faster early; serial is catching up.
- **C_same – B** is zero until $\sim 20\text{M}$ tokens, then settles near -0.028 . The 25M pilot stopped just before it appeared.
- **C_matched – C_same** $\approx +0.027$ throughout: the cost of narrowing every main expert by 20%.
- **C_matched – B** = cost + gain $\rightarrow$ crosses zero at $\sim 67\text{M}$ tokens.

At equal compute, fusion only trades expert width for extra MoE layers: **no net gain** at 100M tokens.

Efficiency: B wins per wall-clock



C_matched: same FLOPs, 9% slower, same loss · C_same reaches B's final loss with ~3% more FLOPs (~5% more time)

Why A and B take the same time

- Same parameters, same FLOPs, same step time (1269.3 ms): **nothing is computed less.**
- PaLM's ~15% speed-up comes from **fusing** attention and MLP input projections, not from overlap.
- Concurrent CUDA streams measured: $\leq 1\%$ — each kernel already fills the A100.
- MoE inputs go through a router and per-expert weights → the PaLM fusion does not carry over directly.

What we can claim

claim	evidence
Parallel $\geq$ serial in this regime	6 conditions, 2 seeds, dense control
Fusion adds ~ 0.03 nats with +15% compute	1 seed, 92% of sequences, below threshold
Fusion gives nothing at equal compute	C_matched = B, 9% slower
Fusion is not shown to be efficient	B best per wall-clock; more tokens likely worth more (~ 0.09 , rough fit)

Limits: 160M active params, 100M tokens (~ 0.6 tokens/param), one shared untuned LR, single-seed MAIN.

Next: Amendment 5 (pre-registered, not yet run)

step	test	cost
1	B_isoflop — B trained on C_same's FLOPs (114.9M tokens). C_same must win by ≥ 0.02 , else stop	~1 h
2	Knock-out of each fusion layer in the trained C_same (no training)	~15 min
3	One fusion layer at the best position vs its iso-FLOP B, 2 seeds	~2.5 h
4	Seed 43 for anything undecided	~1.5 h

Optional: progressive growth (insert fusion layers at 25% of training) if step 1 favours C_same.

Thank you

Code, logs, reports and this deck:

[**github.com/emirkocSU/Transformer-Parallel-Fusion-MoE**](https://github.com/emirkocSU/Transformer-Parallel-Fusion-MoE)

Every number in this deck is read from machine-generated reports in `results/`.